NodePulse

Health monitoring for Portaldot nodes with verifiable on-chain uptime attestations

Portaldot Hacker House · Track: Node Operations & Tooling

`github.com/ifeohsun/portaldot-nodepulse`

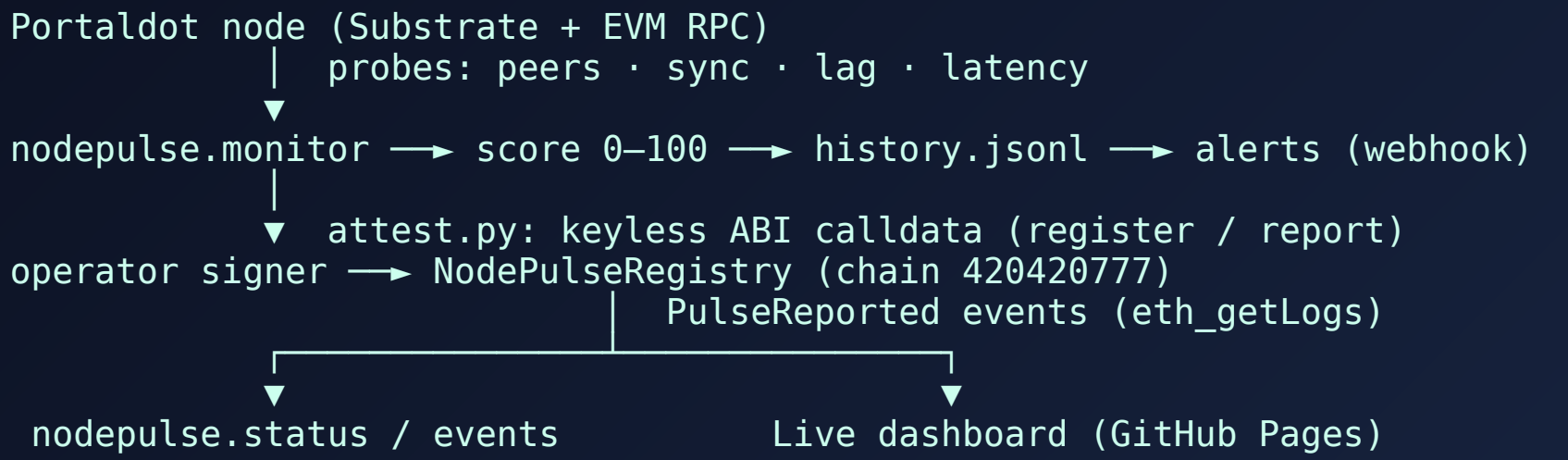
Problem

- Node operators and dApps can't easily tell if a Portaldot RPC / validator node is healthy
- Uptime claims are self-reported and unverifiable
- Outages are discovered by users, not by operators
- No shared, public record of node health on Portaldot

Solution

- Monitor Substrate + EVM JSON-RPC: peers, sync, block lag, latency → 0–100 health score
- JSONL history, uptime stats, watch loop, degrade/recover alerts (Discord webhooks)
- **NodePulseRegistry** contract: operators register nodes and post signed report() attestations
- Backend-free live dashboard reads PulseReported events straight from testnet RPC

Architecture



On-chain proof

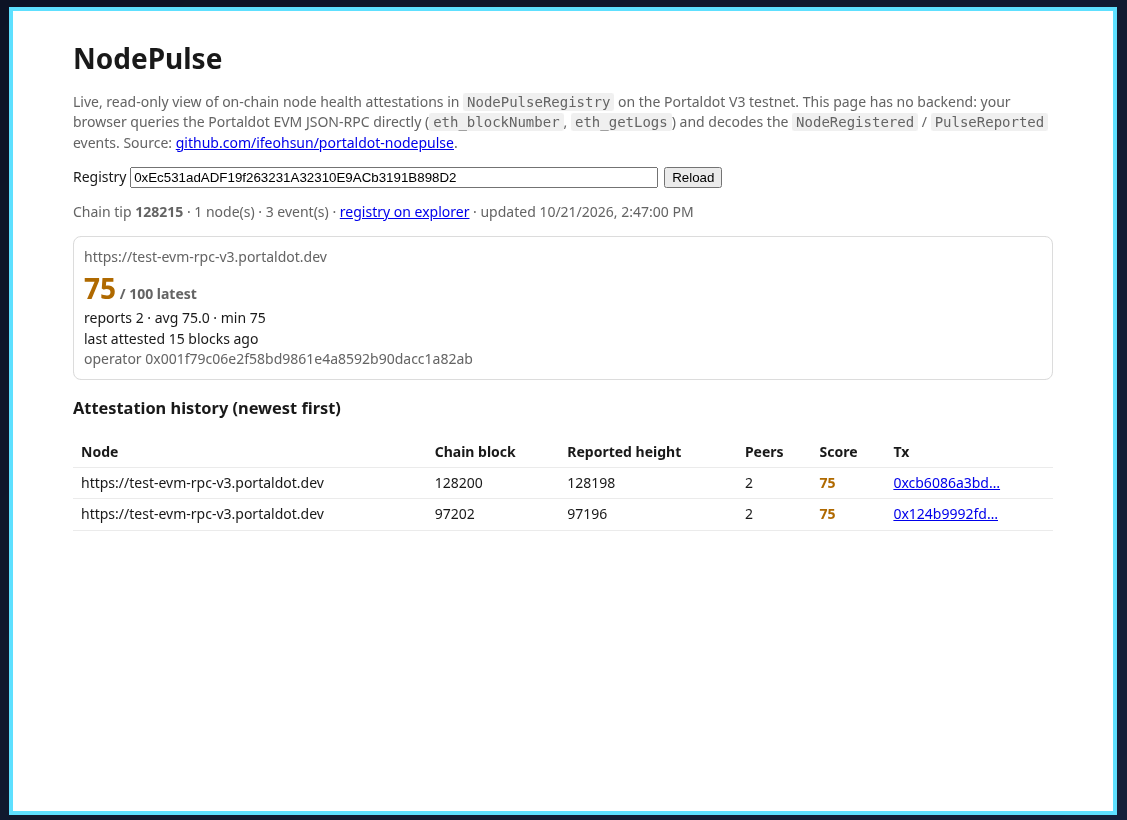
Contract `0xEc531adADF19f263231A32310E9ACb3191B898D2`
Portaldot V3 testnet · EVM chain id 420420777

deploy	`0x952dc3d097c46555e67067ea13232488bc38182862469a2d79fa3ab2ec16708b`
register()	`0x397257e3dfe3a174ec11f6845a98bb7206c4569c674d2af9227f9211bc028c57`
report() #1	`0x124b9992fdda1c384b841e5dda65fc015bfe58fbefd6c090d05b1c9dc2e664fd`
report() #2	`0xcb6086a3bddb22f187ae00e9d201697b5f5d794e750a7d9255171500d16b1970`

Explorer: node-console.portaldot.dev

Demo

Live dashboard: **ifeohsun.github.io/portaldot-nodepulse**



NodePulse

Live, read-only view of on-chain node health attestations in `NodePulseRegistry` on the Portaldot V3 testnet. This page has no backend: your browser queries the Portaldot EVM JSON-RPC directly (`eth_blockNumber`, `eth_getLogs`) and decodes the `NodeRegistered` / `PulseReported` events. Source: github.com/ifeohsun/portaldot-nodepulse.

Registry 0xEc531adADF19f263231A32310E9ACb3191B898D2 Reload

Chain tip **128215** · 1 node(s) · 3 event(s) · registry on explorer · updated 10/21/2026, 2:47:00 PM

https://test-evm-rpc-v3.portaldot.dev

75 / 100 latest

reports 2 · avg 75.0 · min 75

last attested 15 blocks ago

operator 0x001f79c06e2f58bd9861e4a8592b90dacc1a82ab

Attestation history (newest first)

Node	Chain block	Reported height	Peers	Score	Tx
https://test-evm-rpc-v3.portaldot.dev	128200	128198	2	75	0xcb6086a3bd…
https://test-evm-rpc-v3.portaldot.dev	97202	97196	2	75	0x124b9992fd…

Demo video: docs/demo.mp4 · 22 unit tests passing

Roadmap

- Worker that signs & posts report() every N minutes via operator key / remote signer
- Multi-node registry: community RPC + validator nodes
- Score history charts, uptime leaderboard
- Substrate-side metrics (validator / session) and network-security checks
- Mainnet deployment + public status API